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SE LAB-1

Use-Case Flow Specification

Use Case: Submit Adoption Application

Primary Actor: Adoption Applicant

Preconditions:

- The applicant has access to the Pet Adoption & Shelter Management System.
- The selected pet is currently available for adoption.
- The applicant has the required personal and contact information.

Postconditions:

- The adoption application is successfully recorded in the system.
- The application is assigned a unique application ID.
- The applicant receives an application confirmation.
- Shelter Staff can view the submitted application.

Main Success Scenario

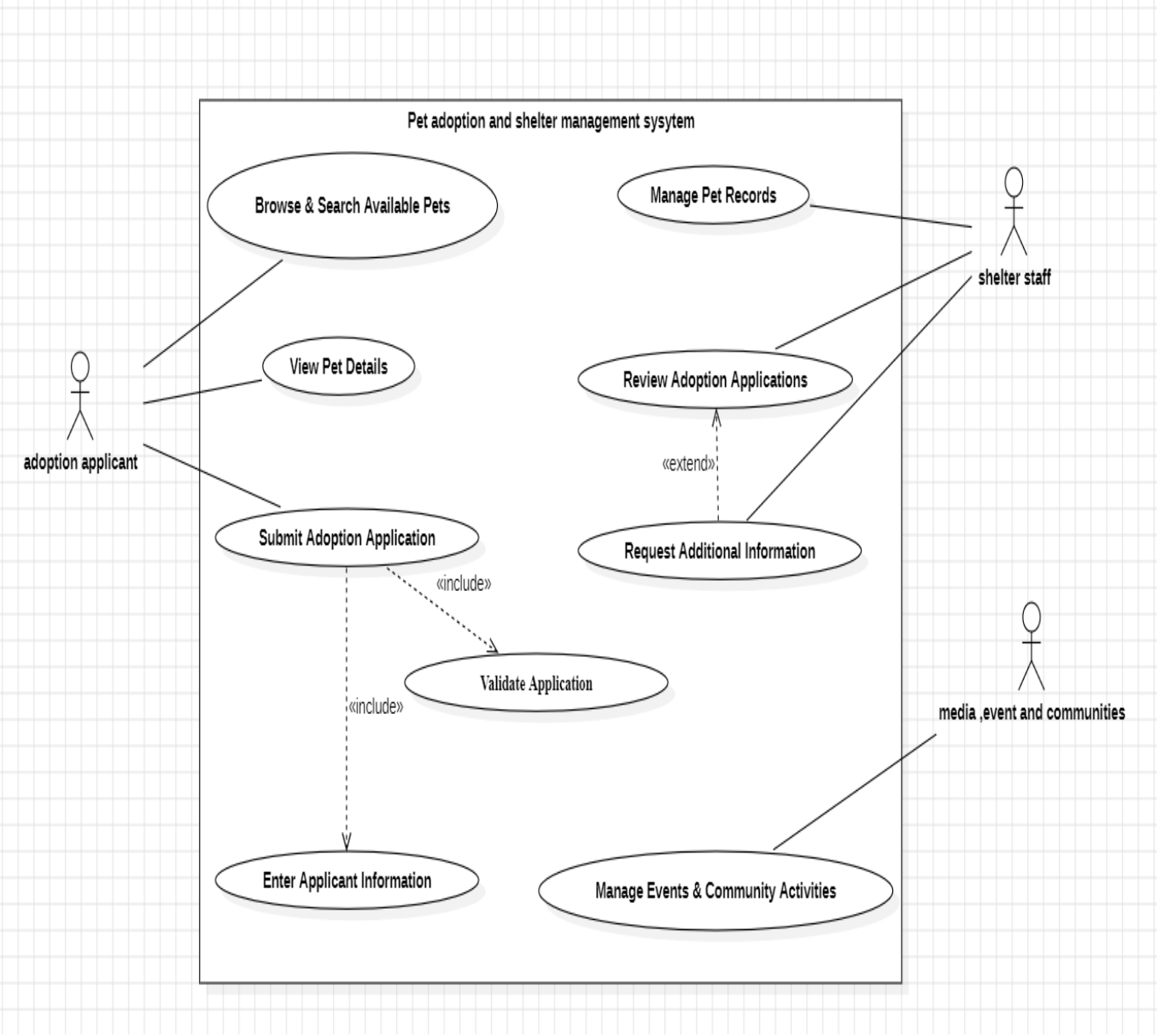
1. The Adoption Applicant browses the available pets.
2. The applicant selects a pet and views its details.
3. The applicant selects **“Submit Adoption Application.”**
4. The system displays the adoption application form.
5. The applicant enters the required personal and contact information.
6. The applicant submits the completed application.
7. The system validates the applicant information.
8. The system records the adoption application.
9. The system generates a unique application ID.
10. The system displays a confirmation message to the applicant.
11. Shelter Staff can view the submitted application for review.
12. The use case ends successfully.

Alternate Flow

A1. Invalid or Missing Information

At Step 7:

1. The system detects missing or invalid information in the application.



2. The system displays an appropriate error message.
3. The applicant is prompted to correct the information.
4. The applicant updates the required fields and resubmits the application.
5. The system validates the information again.
6. If the information is valid, the process continues from Step 8 of the Main Success Scenario.
7. If the applicant does not correct the information, the application is not submitted.

Requirement traceability table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall allow adoption applicants to browse and search available pets using filters such as species, breed, age, and gender.	High	When an applicant applies filters and clicks “Search,” the system displays only pets that match the selected criteria and are available for adoption.	Enables applicants to quickly find suitable pets and improves user experience.	
FR-002	Functional	The system shall allow adoption applicants to submit an adoption application for an available pet.	High	When the applicant submits the application form, the system validates the data, saves the application, and displays an application confirmation with a unique ID.	Core functionality for processing adoption requests.	
FR-003	Functional	The system shall allow shelter staff to add, update, and remove pet records including pet details and adoption status.	High	When staff adds or updates a pet record, the changes are saved and reflected in the pet list immediately.	Ensures accurate and up-to-date information about pets.	
FR-004	Functional	The system shall allow shelter staff to review, approve, or reject adoption applications.	High	When staff approves or rejects an application, the status is updated and the applicant is notified via email/notification.	Critical for managing applications and ensuring transparency.	

NFR-001	Nonfunctional	The system shall respond to pet search requests within 3 seconds under normal load.	High	Performance testing shows that 95% of search requests are completed within 3 seconds.	Ensures good performance and user satisfaction.	
NFR-002	Nonfunctional	The system shall ensure that only authorized shelter staff can modify pet records and adoption application statuses.	High	When an unauthorized user attempts to modify data, the system denies access and logs the attempt.		